Postgres and pgAdmin Setup

DATA SCIENCE BOOTCAMP

NASHVILLE SOFTWARE SCHOOL

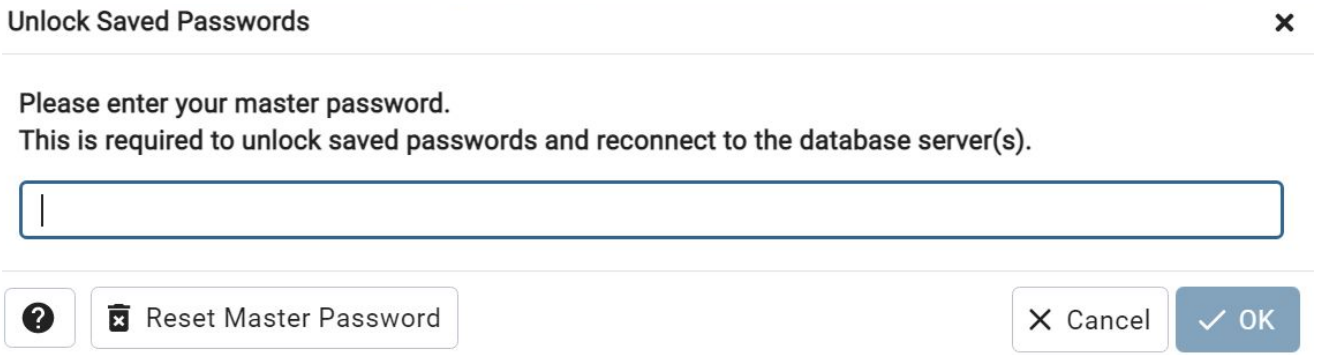
Install PostgreSQL and pgAdmin

- **Important:** We will not be installing the latest version of postgres!
- Download and install the **version 17.8** of PostgreSQL for your OS: https://www.enterprisedb.com/downloads/postgres-postgresql-downloads
  - Linux users will need to use the following link: https://www.postgresql.org/download/.
- During installation you can accept all the defaults **except** you can uncheck **Stack Builder** on the **Select Components** menu
- When asked for a database password, use **postgres**
- Confirm the port number is **5432**
- If pgAdmin was not also installed, you can download the installer from https://www.pgadmin.org/download/.

*Note: Since this is not secure data, we can set our passwords to **postgres**. This is not something you would do with secure data in a production environment*

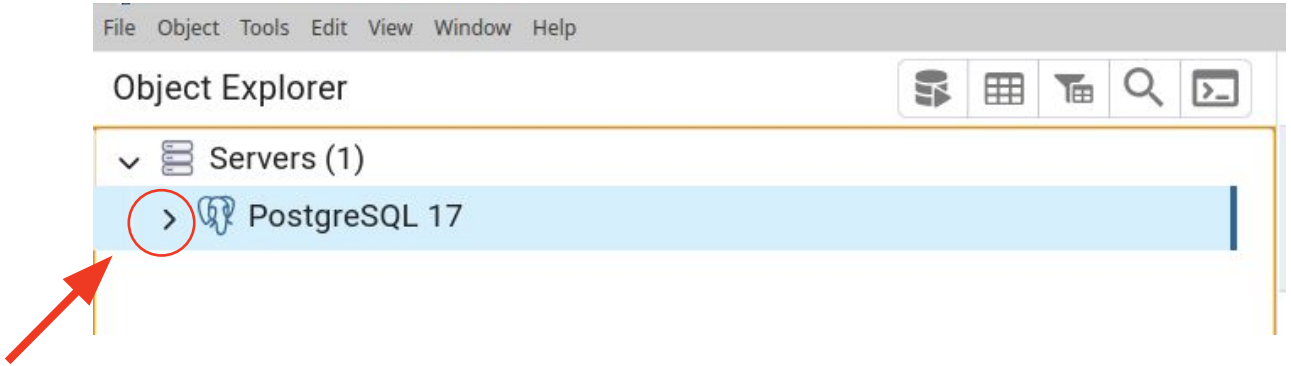
Launch pgAdmin

- Either search for **pgAdmin** in the search bar or navigate to it in your file browser
- Once launched, it will prompt you for a password, which you should have set to **postgres** during the install.



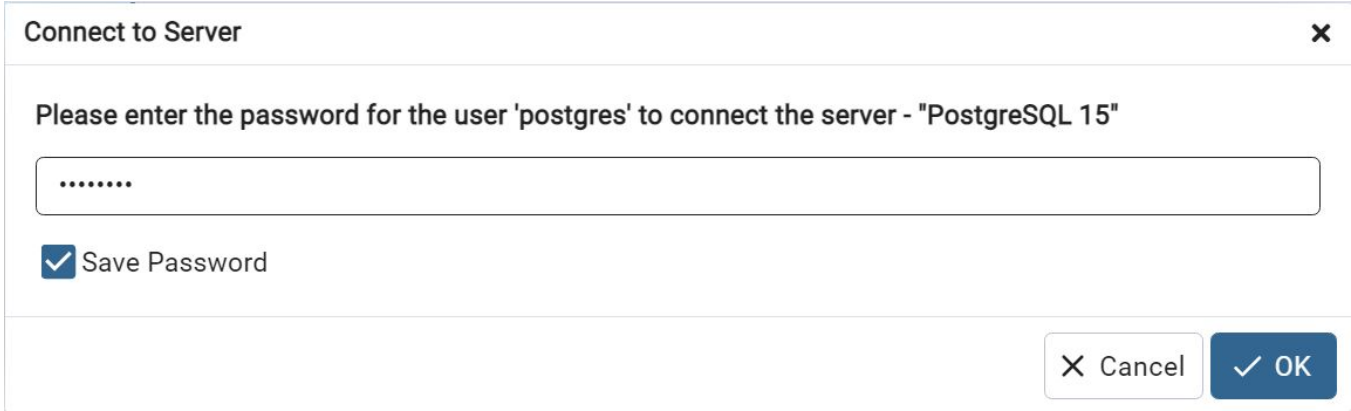
Access Server

- Click the arrow next to **PostgreSQL 17** in the left hand menu



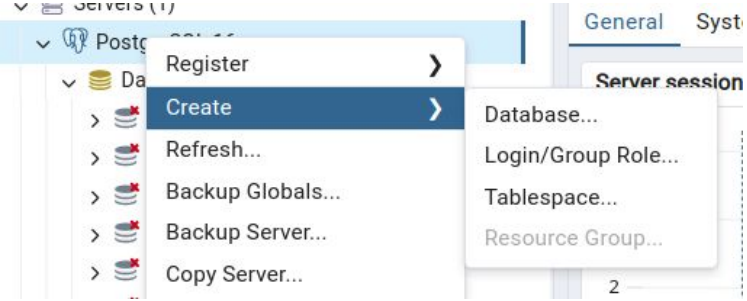
Access Server

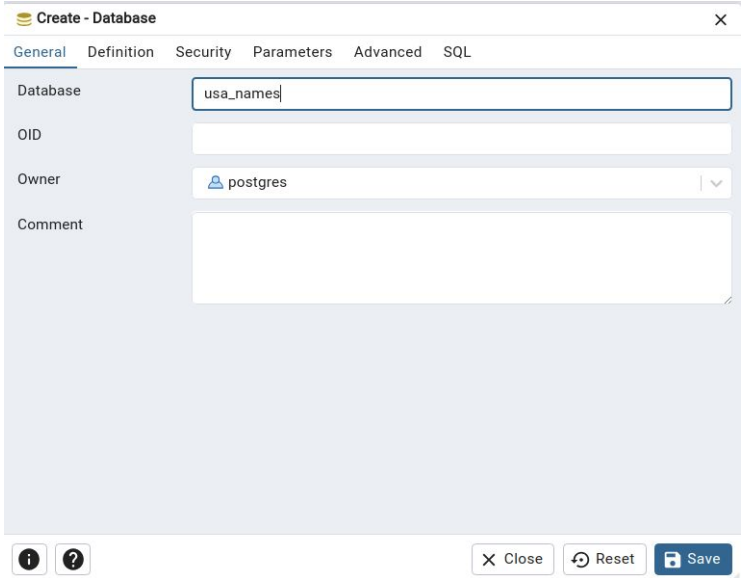
- You will be prompted to enter a password for the server, which is also **postgres**
- Check the **Save Password** box so you won't be asked for a password at this stage again
- Click **OK**



Create a Database

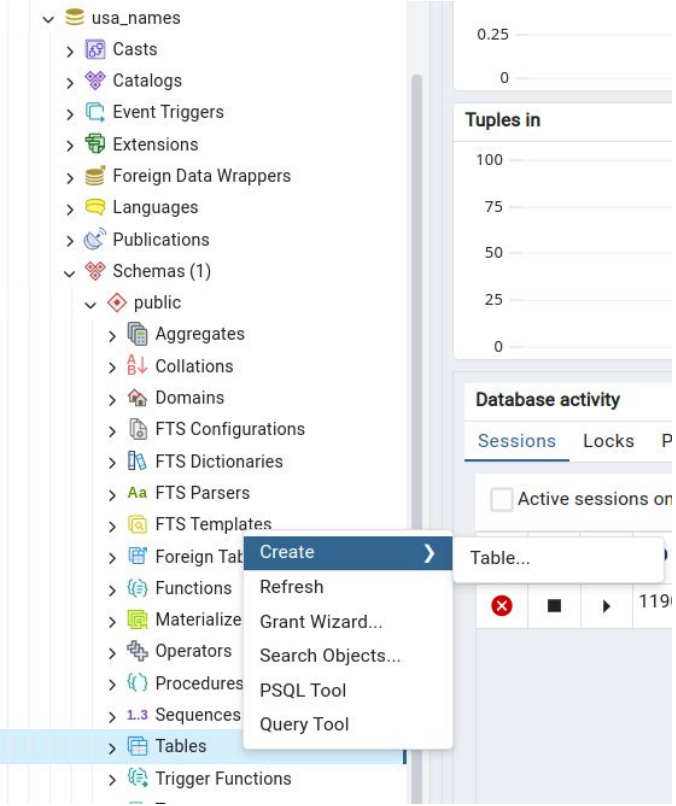
- Right-click on your Postgres Server and choose Create > Database.
- Name this Database “usa_names”.





Create a Table

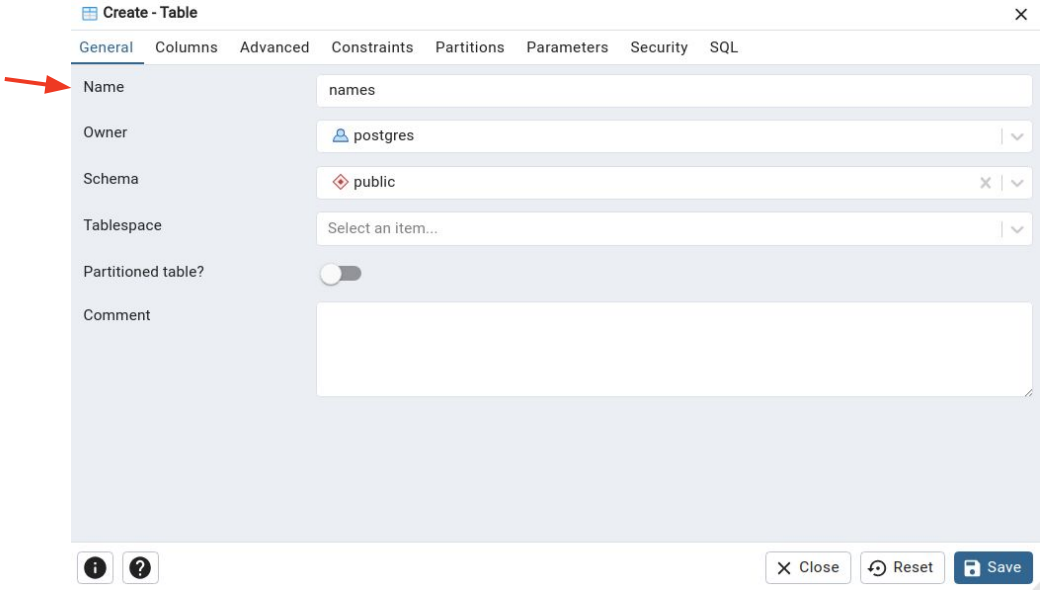
- Download the data located here.
- Expand the usa_names database, right-click the public Schema, and choose Create > Table.



NASHVILLE SOFTWARE SCHOOL

Create a Table

- In the **General** tab, name the table **names**



Create a Table

- Under the **Columns** tab define your columns with the **names** and **data types** shown on the next slide
  - The names need to match what you see **EXACTLY**; be careful not to put in any extra spaces at the beginning or end
  - **Click the + symbol** in the upper right hand corner of the window to create a new column
  - The only fields you should have to edit are **Name** and **Data type**
  - These column names and types correspond to the columns in the csv we will be using
- Click **Save** when you are done

Create a Table

Create - Table

General | Columns | Advanced | Constraints | Partitions | Parameters | Security | SQL

Inherited from table(s): Select to inherit from...

Columns

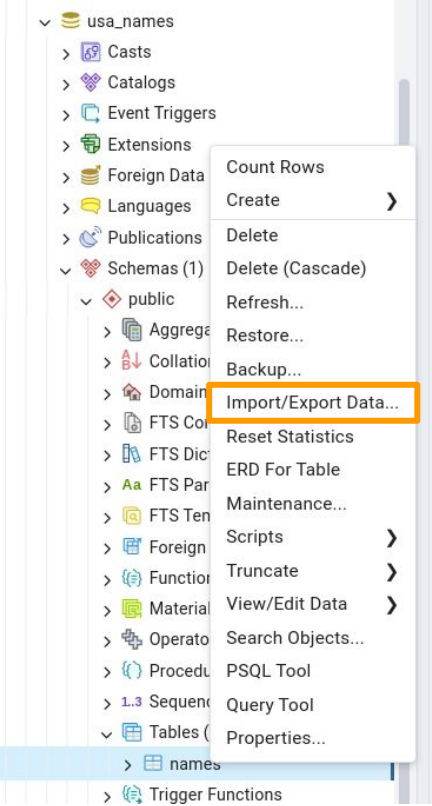
Name	Data type	Length/Precision	Scale	Not NULL?	Primary key?	Default
name	text					
gender	character					
num_registered	integer					
year	integer					

Close | Reset | Save

Create a Table

After clicking Save…

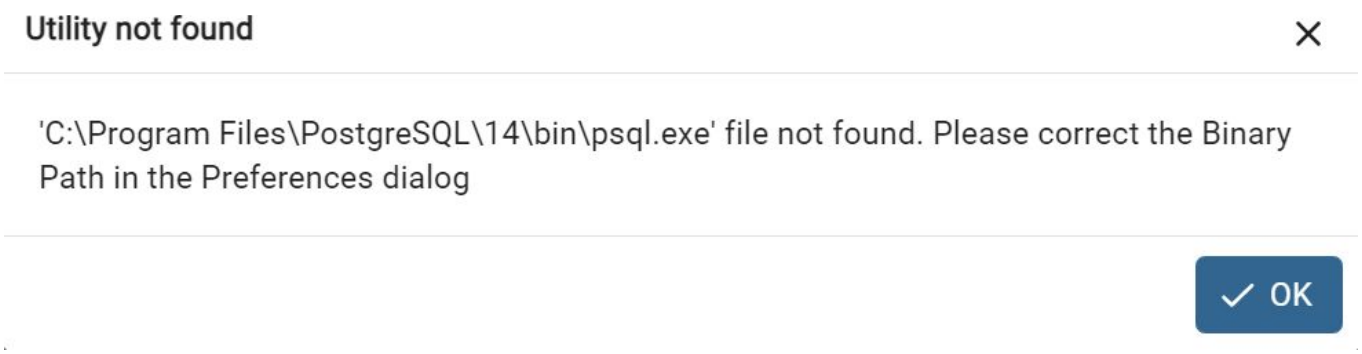
- Right-click on the **names** table
- Select **Import/Export Data…**



NASHVILLE SOFTWARE SCHOOL

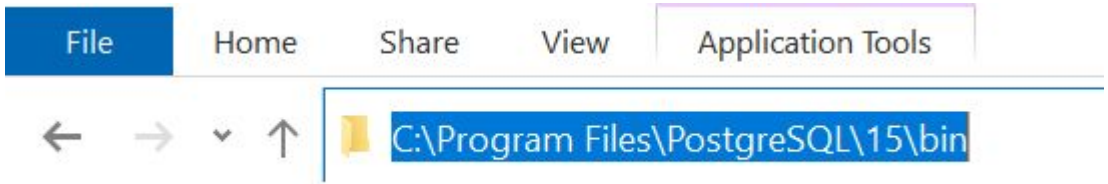
Utility not found

- If you get this error message (or anything similar that mentions the Binary Path), you must take extra steps to set the binary path.



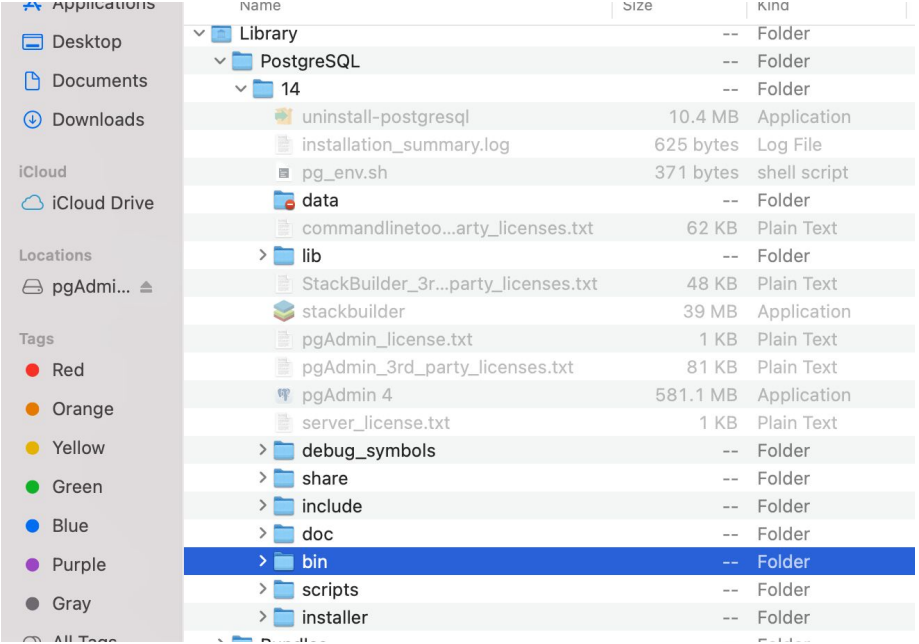
Utility not found

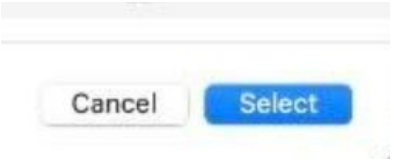
- On Windows, navigate to the **Program Files** folder
- Find the **PostgreSQL** folder and inside. There should be a **16** folder with a **bin** folder inside of that
- Click in the address bar and copy the **entire path** once you are inside the **bin** folder
- It should look something like this:



Utility not found

- On Mac, navigate to the **Macintosh HD** folder
- From there go to **Library → PostgreSQL → 17** and select **bin**
- Click **Select** in the bottom right corner

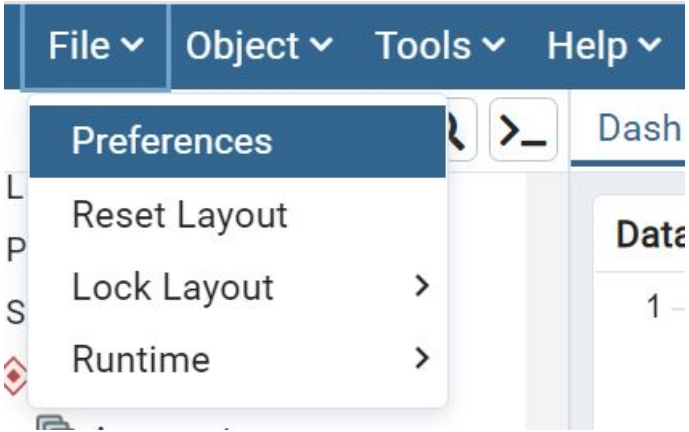




NASHVILLE SOFTWARE SCHOOL

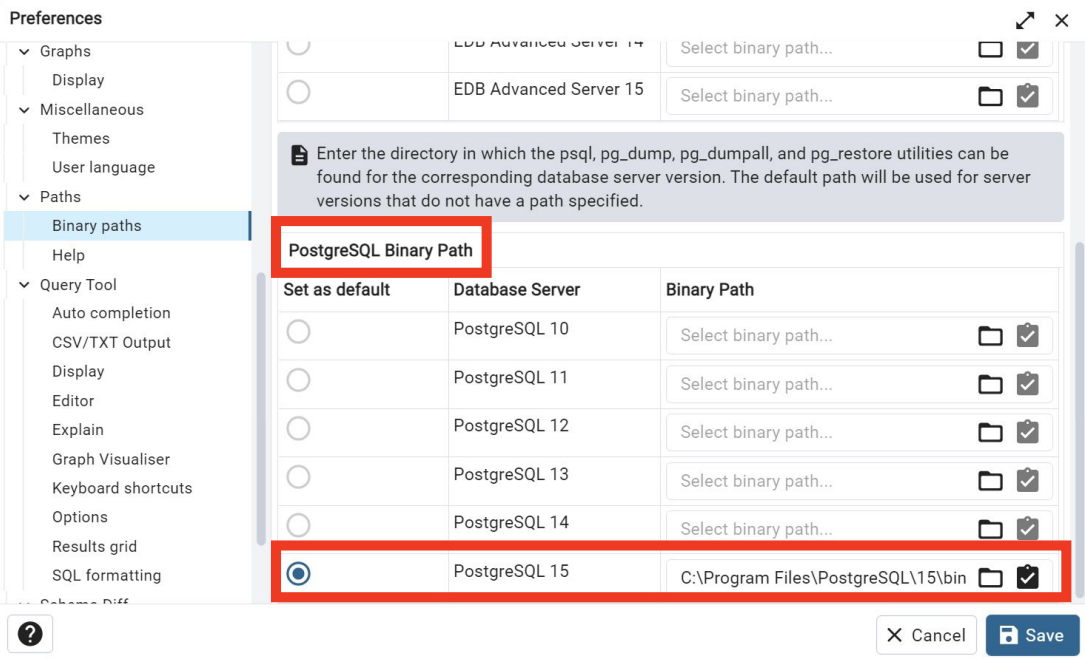
Utility not found

- In pgAdmin, Go to **File** → **Preferences**



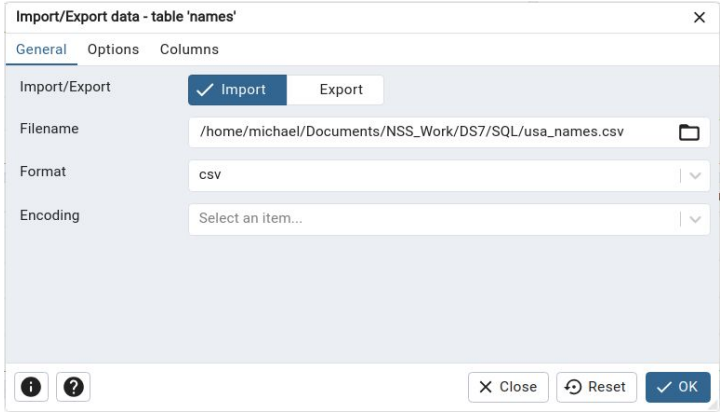
Utility not found

- In the left hand menu, scroll down to **Paths** and click on **Binary Paths**
- Scroll down to the second set of servers called **PostgreSQL Binary Path**
- Paste the path into the **Binary Path** space next to **PostgreSQL 17**
- Check the button next to it under **Set as default**
- Click **Save**



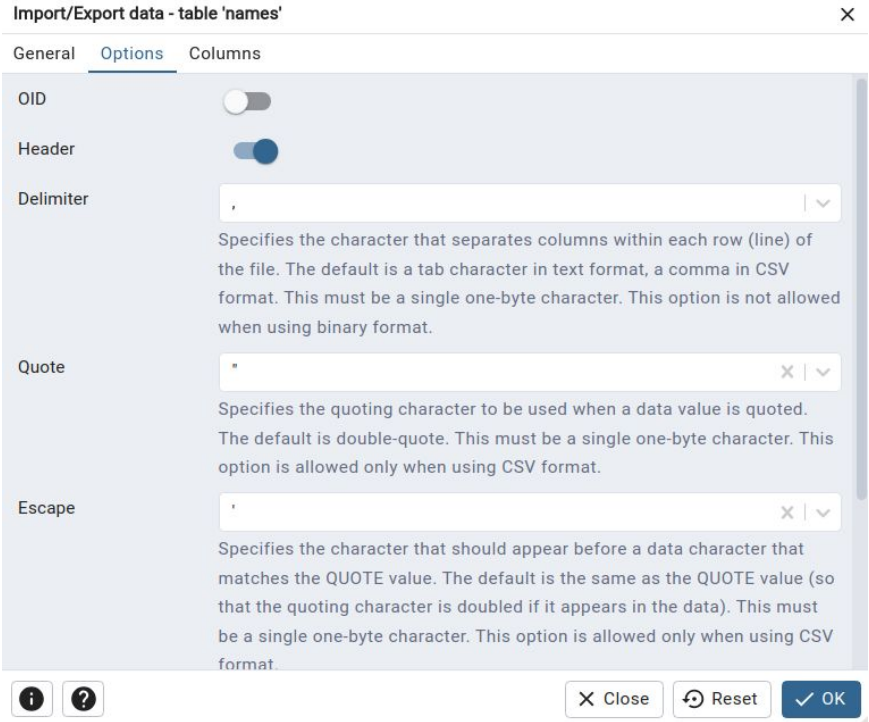
Create a Table

- In the **General** tab, be sure the **Import/Export** option is toggled to **Import**
  Selecting Export will clear out the contents of the CSV and you will have to clone down the data again
- Click on the **folder** icon and navigate to the file **usa_names.csv** in the data folder of the assignment repo
- Double click on the file



Create a Table

- In the **Options** tab, change the **Header** button to **blue** to indicate our file has headers for the columns
- Be sure the **Delimiter** is set to “,”
- Click **OK**



Success!

- Your data will import and show the following success message

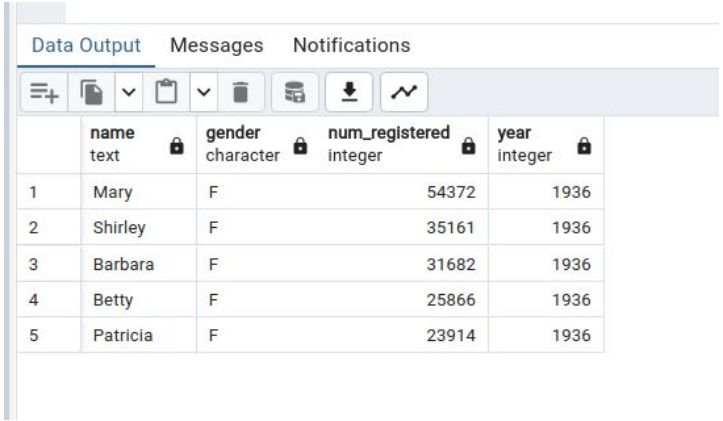
Process completed

Copying table data 'public.names' on database 'usa_names' and server 'PostgreSQL 16 (localhost:5432)'

View Processes

Verify the table was created

- Right-click on the usa_names database and select Query Tool
- Input the query “SELECT * FROM names LIMIT 5;” and click the Execute Script button (the triangle) at the top of the screen, and verify that you get output in the bottom part of the screen.



Data Output | Messages | Notifications

	name text	gender character	num_registered integer	year integer
1	Mary	F	54372	1936
2	Shirley	F	35161	1936
3	Barbara	F	31682	1936
4	Betty	F	25866	1936
5	Patricia	F	23914	1936